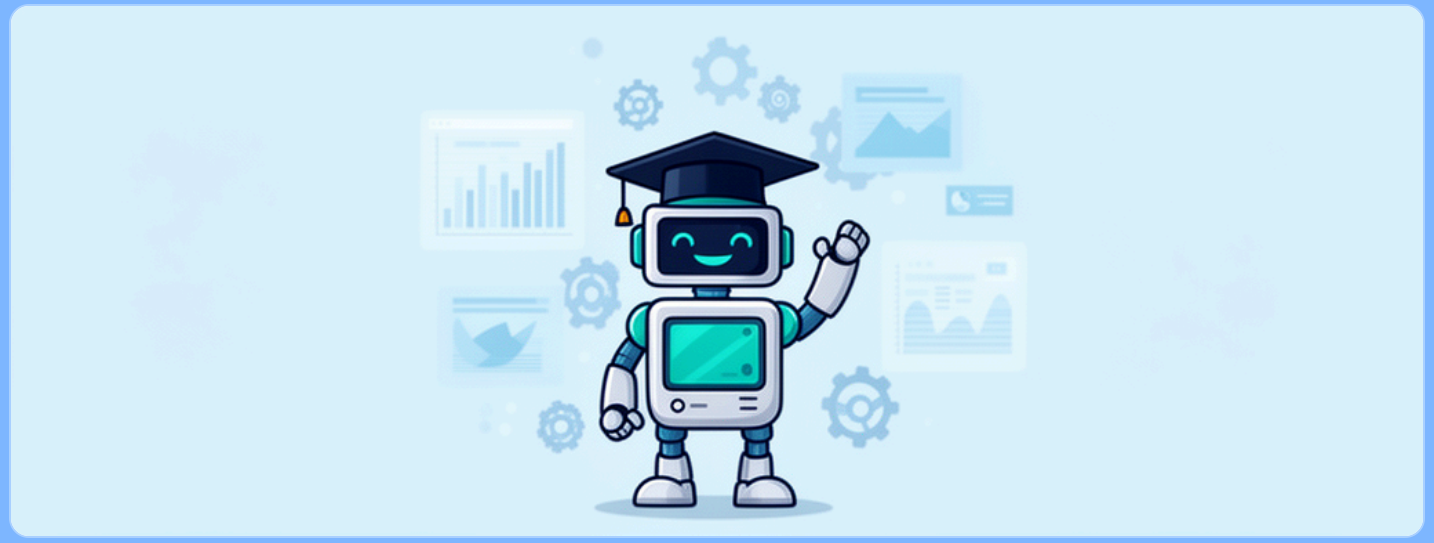
The 20-Questions Machine

Day Two — You Draw a Decision Tree

Block 1 · Session 2

.By the end of today, YOU will draw a decision tree by hand



The Game of 20 Questions

.I'm thinking of an animal

.You may only ask **yes/no** questions

...The counter climbs



You Were a Tree Just Now

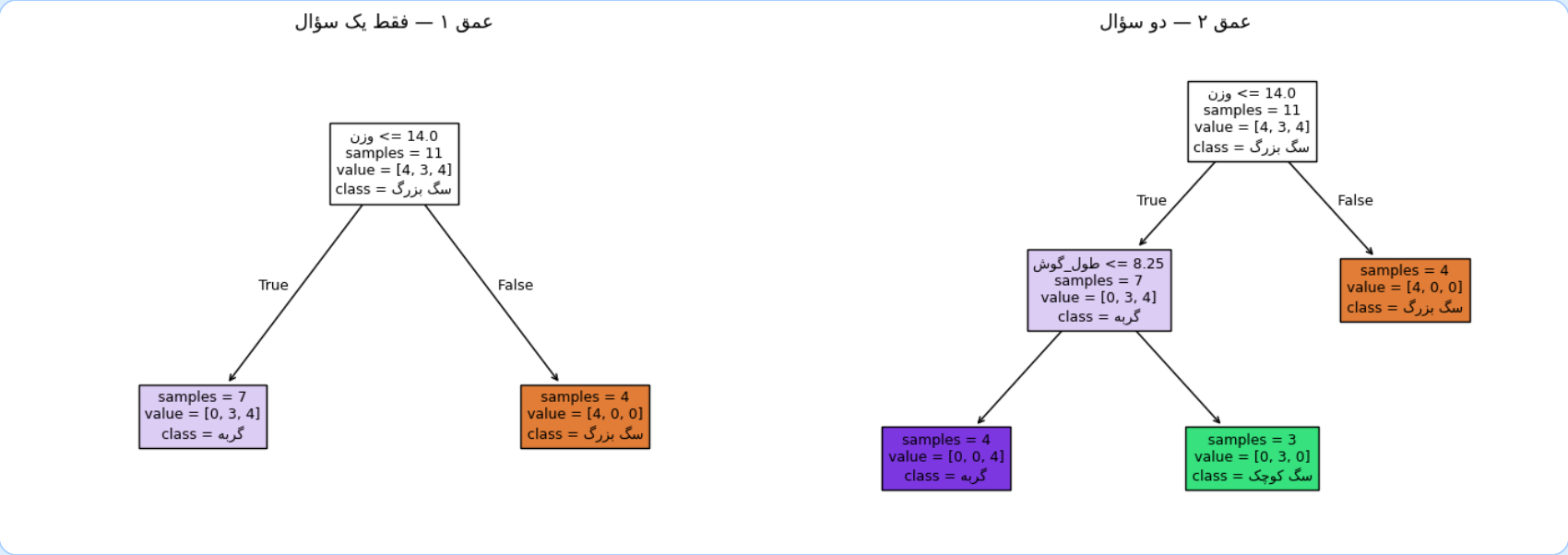
.Every answer was a **door** — every door, a new path

.This shape is **a decision tree** — Session 1's machine plays it too

Now Let the Machine Play

!Last session: one question ← the puppy became a cat

.Today: it is allowed **two questions**

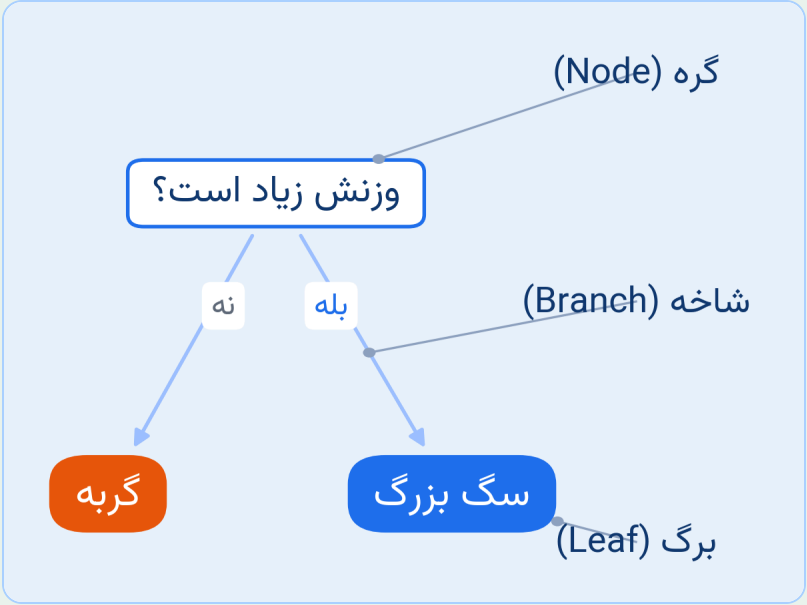


Tree Anatomy

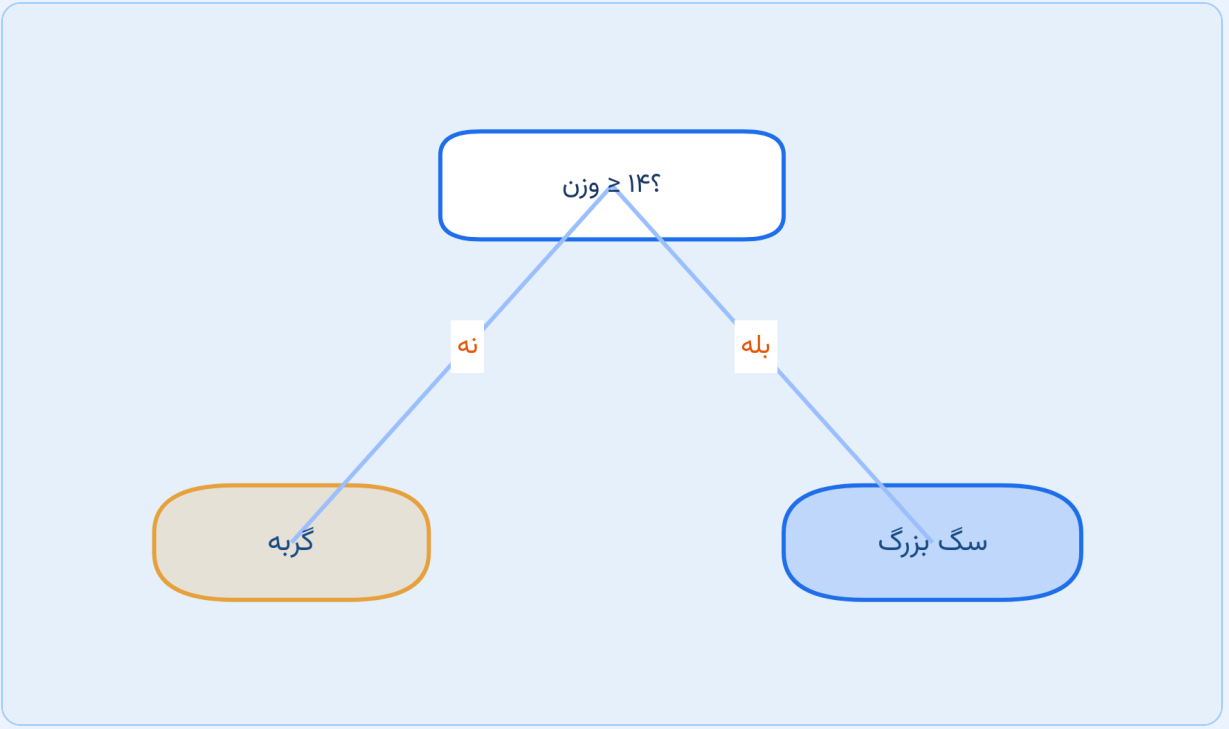
Node = a question — a fork with a signpost

Branch = one answer's door (yes/no)

Leaf = the final room — where the verdict is said



Session 1's Machine = One Question



.Last session's red line, drawn as a tree

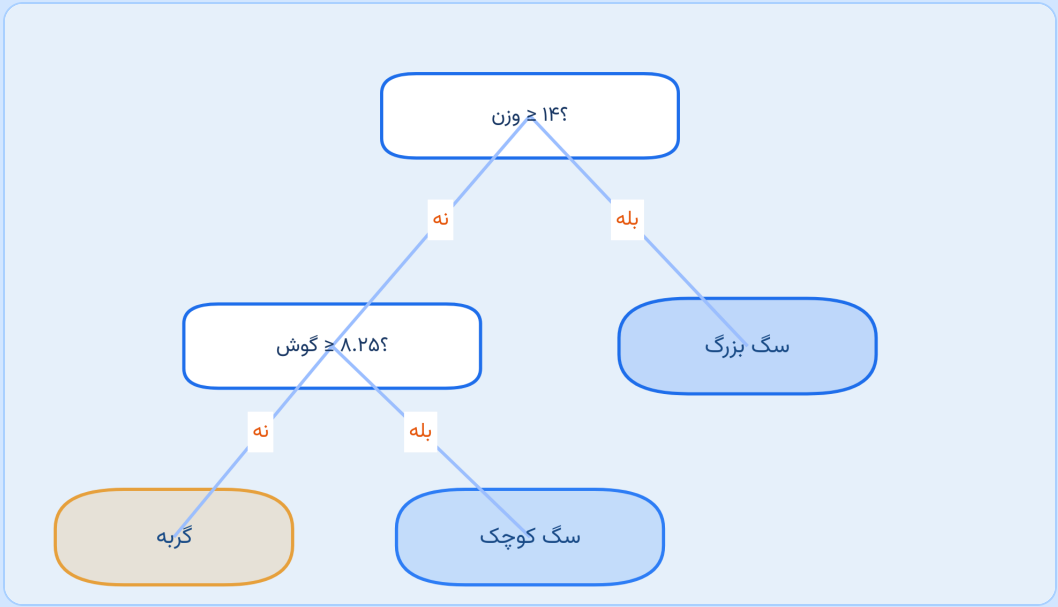
.One question, two doors, two verdicts

The Root: The First Question

. `weight >= 14` The machine chose

?What did YOU ask first in the game

Two Questions, Three Rooms



.Big dogs with **one** question; cats and small dogs need the second

?Which Door Does the Elephant Take

yes ← `weight >= 14?`

!Verdict: big dog — with no second question

Read the Path: puppy

no ← `weight >= 14?`

yes ← `ears >= 8.25?`

!Verdict: small dog — correct

The Depth Knob (max_depth)

the maximum number of questions = `max_depth`

Depth 1: the puppy becomes a cat — 73% correct

Depth 2: everything correct — 100%

Put Your Hand on the Knob

~/ml-course

```
max_depth = 2    # change the number

model = DecisionTreeClassifier(max_depth=max_depth)
model.fit(X, y)
```

?What is the smallest perfect depth

Lab: Draw Your Own Tree

.animals. Three rooms. Yes/no questions only 11

.Paper and pencil. Any tree that works, wins



The Tree Battle

.Swap papers

.Surprise animals: **rabbit. fox. hamster. sparrow. lion**

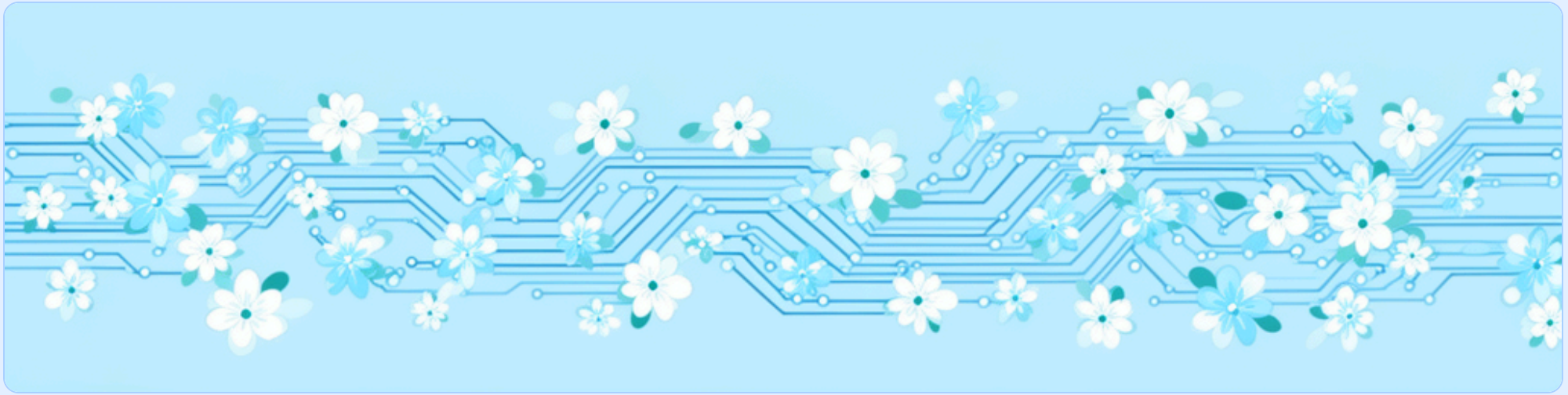
!Funniest failure gets applause

Your Tree = The Machine's Tree

.Your paper tree and `plot_tree` are the same shape

.Question, door, question, door, verdict

.Tonight: commit a photo of your tree



What You Learned Today

- A tree = **node, branch, leaf**
- **Root** = the first question; **depth** = the most questions
- Depth 2 cured the puppy mistake
- Different trees can all be correct

Tonight's Delivery

A photo of your paper tree ← `s02_tree.jpg` ← your GitHub repo

Three small notebook problems: count, predict, find the right depth

At-Home Problems (3)

B1 The question and leaf counter .1

?B2 Which room does the elephant reach .2

B3 The smallest perfect depth .3

instant score, anytime ← `python judge/run.py s02_notebook.ipynb`

Next Session

?`weight >= 14` Where did the machine get

?WHO chooses the questions

.Next time: you become the question-picker

